

Automated MoCA Score Estimation Using Eye-Gaze Data and Vision Transformers

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Abstract—Cognitive impairment is a growing public health concern, with early detection playing a crucial role in improving patient outcomes. The Montreal Cognitive Assessment (MoCA) is widely used for screening mild cognitive impairment (MCI) and early-stage dementia. However, traditional MoCA assessments require manual scoring by trained professionals, making the process labor-intensive, time-consuming, and susceptible to human error. To overcome these limitations, we propose an automated pipeline for MoCA score estimation using eye-gaze data and Vision Transformers (ViTs). Our approach leverages gaze-tracking technology to capture spatial and temporal eye-movement patterns during structured cognitive tasks, identifying subtle cognitive impairments that may otherwise go unnoticed. The raw gaze data is preprocessed and mapped onto task-relevant image regions, where a pretrained ViT extracts high-dimensional feature representations. To address inconsistencies in gaze sampling and improve temporal modeling, we introduce a time-aware positional embedding mechanism that enhances the model’s ability to infer cognitive performance. These extracted features are then processed by a transformer-based classification model to predict MoCA scores with high accuracy. We validate our approach using a dataset collected from seven cognitive gaming sessions, demonstrating its effectiveness in automated cognitive assessment. The experimental results indicate that our method provides a reliable and efficient alternative to traditional MoCA evaluations, reducing dependency on human intervention while maintaining diagnostic accuracy.

Index Terms—Deep Learning, Vision Transformer (ViT), Eye-Gaze Tracking, Mild Cognitive Impairment (MCI)

I. INTRODUCTION

Cognitive impairment is a significant public health concern, with early detection playing a crucial role in managing and mitigating its effects. The Montreal Cognitive Assessment (MoCA) [1] is a widely used screening tool for detecting mild cognitive impairment (MCI) and early-stage dementia. Traditionally, MoCA scoring relies on direct assessments, requiring manual evaluations by trained professionals. However, manual processing of cognitive assessment data is labor-intensive, time-consuming, and prone to errors, particularly when dealing with large datasets or complex multi-modal inputs such as eye-

gaze data. Automating this process can enhance efficiency, reduce bias, and enable scalable, objective assessments [2].

Eye-gaze tracking has emerged as a promising non-invasive method for cognitive assessment, capturing subtle impairments in attention, visual processing, and executive function. In this study, we explore the feasibility of estimating MoCA scores based on eye-gaze data gathered during a structured seven-game experiment designed to engage various cognitive functions. The recorded gaze patterns provide rich temporal and spatial information about participants’ cognitive processing, making them a valuable but complex data source for automated analysis. However, conventional approaches for processing such data typically rely on manual feature extraction and heuristic-based analysis, which require significant effort and introduce inconsistencies due to human subjectivity [3]. These limitations highlight the need for a more efficient, automated pipeline that can process gaze data reliably and at scale.

To address these challenges, we propose a processing pipeline that leverages a pretrained Vision Transformer (ViT) model for automated feature extraction and MoCA score prediction. The pipeline involves pre-processing raw eye-gaze data, extracting meaningful representations using ViT, and mapping these features to cognitive performance indicators. To evaluate its effectiveness, we tested the pipeline on a dataset acquired during an experiment containing seven cognitive gaming sessions, demonstrating its potential for automating MoCA score estimation. By leveraging deep learning transformer-based techniques, our approach eliminates the need for manual feature engineering, enabling a more robust and scalable method for cognitive assessment.

II. RELATED WORK

Eye-tracking has gained attention as a non-invasive method for evaluating cognitive function, particularly in aging populations and individuals at risk of mild cognitive impairment (MCI) and Alzheimer’s disease (AD)[4]. Eye movement metrics, such as fixation duration, saccadic amplitude, and pupil dilation, provide insights into attention, memory, and executive functioning. Lagun et al. [5] demonstrated that gaze-based features could distinguish between healthy individuals and

those with MCI with high accuracy. Similarly, in [6], the authors reviewed how abnormalities in saccadic behavior are linked to cognitive decline in neurodegenerative diseases.

Several studies have also explored the integration of eye-tracking data with cognitive tasks to enhance MCI screening. Hicks et al. [7] used an eye-tracking version of the Trail Making Test (TMT) to predict cognitive impairment, showing strong correlations between gaze patterns and clinical cognitive scores. Additionally, Jang et al. [8] proposed a multimodal approach combining eye-tracking and speech features to classify MCI, demonstrating the effectiveness of gaze analysis in remote cognitive assessment.

MoCA is one of the most widely used screening tools for mild cognitive impairment and early dementia detection [9], it evaluates multiple domains, including attention, visuo-spatial skills, memory, and executive functioning. Although MoCA is traditionally administered by clinicians, recent research has investigated the use of machine learning to automate cognitive screening. In [10] for instance, the authors employ Random Forests to predict MoCA scores based on a set of social and biomarker data. However, feature selection (as in almost all cases of machine learning approaches, except deep learning) was made manually, and this adds to the complexity of experiment execution. In [11] the authors explored the possibility of processing mechanical and touchscreen keyboard strokes with machine learning methods to predict MoCA scores. Their approach relies on handcrafted features extracted from keystroke dynamics and introduces a clinical outcomes framework that maps multiple neuro-psychological scales into cognitive subdomains.

With the growing interest in remote and automated cognitive testing, researchers have explored alternative modalities, such as eye-tracking and speech analysis, to estimate MoCA scores without direct clinical supervision. Both supervised [5] and unsupervised learning [12] approaches applied on eye-gaze patterns during cognitive-related tasks demonstrated their capability in MoCA score approximation, reinforcing the feasibility of gaze-based cognitive screening. Deep learning has significantly advanced the field of cognitive impairment detection, particularly in analyzing complex behavioral and physiological signals. Recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) networks have been widely applied to process sequential gaze data. LSTMs are also used in a personalized framework to predict mental states from eye-tracking data during video viewing, achieving improved accuracy over traditional machine learning models [13]. However, this approach targets transient affective states (e.g., stress vs. calm) rather than structured cognitive assessment, and it operates on low-level gaze features without leveraging visual context or high-level spatial representations as in transformer-based architectures. Additionally, convolutional neural networks (CNNs) and Gated Recurrent Units (GRUs) have been utilized to extract high-level features from gaze heatmaps and eye-tracking videos for cognitive analysis. For example, in [14] the authors proposed VTNet, a dual-branch architecture combining CNNs and GRUs to process visual

scan-paths and temporal gaze sequences in parallel, achieving promising results in early Alzheimer’s detection. Unlike our method, which employs a unified Vision Transformer architecture for joint spatiotemporal modeling, their approach relies on task-specific feature fusion and separate processing streams, potentially limiting generalizability across different tasks or gaze conditions. Transformers have revolutionized deep learning by leveraging self-attention mechanisms to capture long-range dependencies in sequential data [15]. While originally designed for natural language processing (NLP), Transformers have been successfully applied to time-series analysis, computer vision, and biomedical data. Recent studies have explored Transformer-based models for cognitive assessment and neurological disorder prediction. Dosovitskiy et al. [16] introduced the Vision Transformer (ViT), demonstrating its capability in visual representation learning. More recently, Mondal et al. [17] proposed an attention-based framework for gaze-driven human-computer interaction, highlighting the applicability of Transformers in eye-tracking analysis.

Given these advancements, transformer-based architectures, and Vision Transformers in particular, hold significant potential for modeling complex gaze dynamics. However, their application to structured cognitive assessment—particularly for estimating standardized cognitive scores like MoCA—remains largely unexplored. In this work, we introduce an end-to-end framework that leverages ViTs to infer cognitive status from eye-gaze data collected during structured tasks, providing an automated alternative to traditional MoCA evaluation.

III. METHOD

A. Overview

We propose a transformer-based architecture for automated cognitive assessment, which predicts MoCA scores using gaze-aligned visual input. The core idea is to process only the image regions where visual attention occurred, thereby reducing input dimensionality and enhancing focus on cognitively relevant content. To further improve temporal modeling, we introduce a novel Time-Aware Positional Embedding (TAPE) mechanism that accounts for non-uniform gaze sampling intervals and encodes temporal attention dynamics. The architecture integrates a pretrained Vision Transformer (ViT) to model both spatial and temporal dependencies in gaze-driven visual input and produces MoCA score predictions via multi-head classification.

B. Gaze-Conditioned Patch Selection

During data collection, participants interacted with a series of structured cognitive games displayed on a tablet device, while their eye-gaze coordinates $\mathbf{g}_t = (x_t, y_t)$ were continuously recorded using an eye tracker mounted directly on the tablet. Simultaneously, screenshots of the tablet interface were captured at a fixed frame rate and timestamped, ensuring temporal alignment between visual stimuli and gaze data.

To construct a compact and informative input sequence, we leverage gaze data to select only the embedded patches corresponding to regions actually attended by the participant.

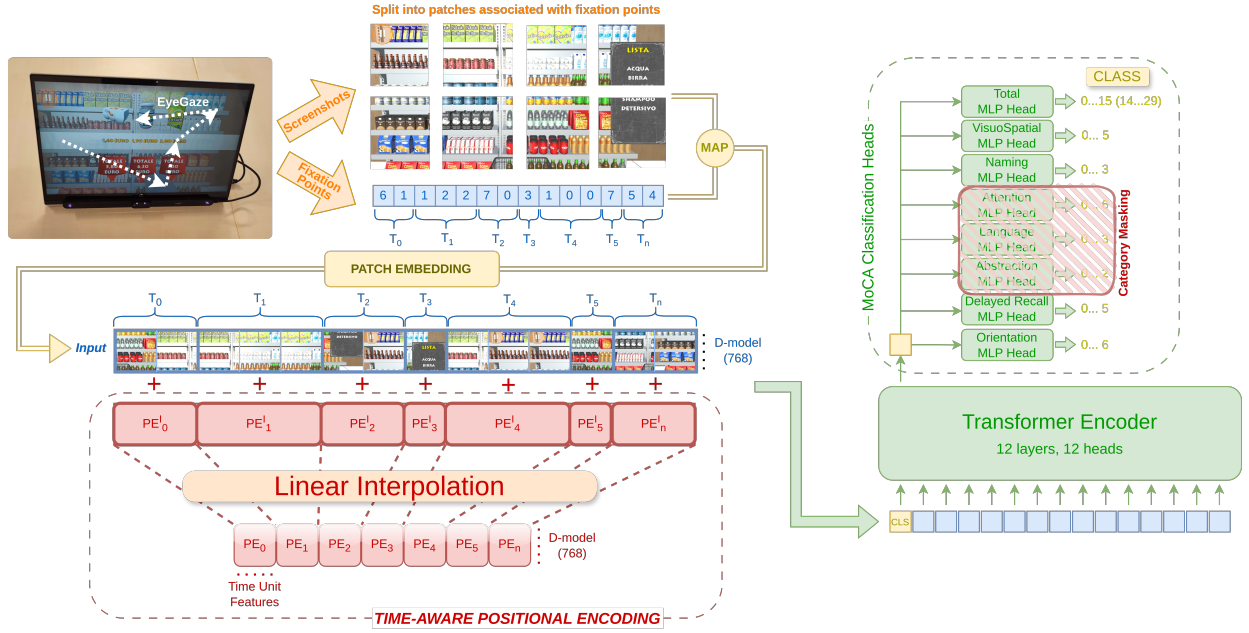


Fig. 1. DA-ViT (Dynamic-Aligned Vision Transformer) architecture. Our major contribution lies in replacing the standard positional encoder with TAPE, which incorporates information about the temporal duration of fixation point. This helps standardize the representation of time, compensating for the highly variable eye-gaze sampling rates, which can differ significantly across consecutive time intervals.

Specifically, we consider fixed-duration input segments of T seconds (e.g., $T = 10$), during which a single game frame $\mathbf{I}$ is displayed. Each frame $\mathbf{I}$ is divided into a non-overlapping grid of $P \times P$ patches, following the standard Vision Transformer (ViT) tokenization scheme. All patches are independently passed through a shared patch embedding layer, producing a spatial token tensor $\mathbf{Z} \in \mathbb{R}^{P^2 \times d}$, where d is the embedding dimension.

Each input segment is temporally partitioned into $N = T/t_w$ non-overlapping windows of duration t_w seconds (e.g., $t_w = 1$), which are used to capture the temporal structure of gaze behavior. For each window $w = 1, \dots, N$, we generate a gaze-driven sequence of patch tokens as follows:

$$\mathcal{S}_w = [\mathbf{z}_1^{(w)}, \mathbf{z}_2^{(w)}, \dots, \mathbf{z}_{K_w}^{(w)}] \quad (1)$$

Here, $\mathbf{z}_k^{(w)}$ denotes the embedding of the k -th patch $\mathcal{P}_k \subseteq \mathbf{I}$ that is intersected by a gaze point at time $t_k \in [(w-1)t_w, wt_w]$, indicating that the gaze event occurred within the w -th temporal window. Thus, each token $\mathbf{z}_k^{(w)}$ represents a viewed patch, and the sequence $\mathcal{S}_w$ preserves the temporal order of the corresponding gaze events.

The window index w serves as a proxy for temporal position and is later used to generate **Time-Aware Positional Embeddings**, allowing the model to incorporate fine-grained temporal context in the encoding of gaze behavior.

This patch selection strategy ensures that only gaze-relevant regions are processed while retaining the spatial fidelity of visual stimuli. By computing patch embeddings independently of gaze data and selecting attended regions only afterward, the model preserves the full visual detail of each frame

while reducing the computational cost of sequence processing. This allows attention to be focused on cognitively meaningful regions without sacrificing representational fidelity.

C. Time-Aware Positional Embedding (TAPE)

Eye-gaze data is often affected by inconsistencies due to both intrinsic factors (e.g., operating system scheduling, CPU load) and extrinsic factors (e.g., inaccurate gaze acquisition caused by improper face positioning or illumination changes). These issues result in variable sampling rates and irregular temporal distributions of gaze points across different sequences.

To incorporate temporal structure into gaze-driven input sequences, we introduce a **Time-Aware Positional Embedding (TAPE)** mechanism that is robust to variable sampling rates in eye-tracking data.

In contrast to standard Vision Transformers, which use a fixed-size learnable positional embedding with length equal to the input sequence, our setting requires a more flexible formulation. Each input segment of duration T seconds is divided into T_u non-overlapping temporal windows of fixed duration t_w seconds (i.e., $T_u = T/t_w$). Each window may contain a variable number of gaze points and, consequently, a variable number of gaze-attended patch tokens.

To model temporal structure across windows, we learn a sequence of positional embeddings $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_{T_u}]$, where $\mathbf{e}_w \in \mathbb{R}^d$ encodes the temporal position of the w -th window. Since the number of attended patches in each window can vary across samples, we interpolate each embedding vector $\mathbf{e}_w$ over the actual number of tokens n_w observed in window w .

Let $\mathbf{e}_w^{\text{interp}} \in \mathbb{R}^{n_w \times d}$ denote the interpolated version of $\mathbf{e}_w$, obtained via linear interpolation over n_w steps. For each patch token $\mathbf{z}_i^{(w)}$ observed in the time window w , the final temporally enriched token becomes:

$$\mathbf{h}_i^{(w)} = \mathbf{z}_i^{(w)} + \mathbf{e}_w^{\text{interp}}[i] \quad \text{for } i = 1, \dots, n_w \quad (2)$$

This strategy allows each patch token to be augmented with a time-aware embedding that is both globally learnable (shared across sequences) and locally adaptable (interpolated to per-sample sequence length), enabling consistent modeling of temporal dynamics despite sampling variability.

The resulting temporally ordered sequence

$$\mathcal{H}_w = \left\{ \mathbf{h}_1^{(1)}, \dots, \mathbf{h}_{n_1}^{(1)}, \dots, \mathbf{h}_1^{(T_u)}, \dots, \mathbf{h}_{n_{T_u}}^{(T_u)} \right\} \quad (3)$$

is then passed to the transformer encoder.

D. Transformer-Based Gaze Encoding

The temporally enriched sequence

$$\mathcal{H}_w = \{ \mathbf{h}_1^{(1)}, \dots, \mathbf{h}_{n_1}^{(1)}, \dots, \mathbf{h}_1^{(T_u)}, \dots, \mathbf{h}_{n_{T_u}}^{(T_u)} \} \quad (4)$$

obtained through the TAPE mechanism, is passed to a ViT to model long-range dependencies across gaze-driven visual tokens.

A learnable classification token $\mathbf{h}_{[\text{CLS}]}$ is pre-pended to the sequence to summarize the global context:

$$\mathbf{H} = [\mathbf{h}_{[\text{CLS}]}, \mathbf{h}_1, \dots, \mathbf{h}_L] \quad (5)$$

where $L = \sum_{w=1}^{T_u} n_w$ is the total number of attended tokens across all windows in the input segment.

This sequence is then processed through the transformer encoder, which applies multiple layers of self-attention and feedforward transformations:

$$\mathbf{H}' = \text{TransformerEncoder}(\mathbf{H}) \quad (6)$$

The output embedding corresponding to the classification token, denoted as $\mathbf{h}'_{[\text{CLS}]} \in \mathbb{R}^d$, serves as a compact representation of the participant's spatiotemporal gaze behavior over the input segment.

This final embedding is then used for cognitive score prediction.

E. MoCA Score Prediction

We formulate MoCA estimation as a multi-head classification task, where each head predicts one of the MoCA categories as well as the total score. Let M be the number of MoCA categories, and let $\hat{\mathbf{y}}_m$ and $\hat{\mathbf{y}}_T$ denote the predicted probability distributions for category m and the total score, respectively.

Each prediction head consists of a fully connected layer followed by a softmax activation. The model is trained by minimizing the sum of categorical cross-entropy losses:

$$L_{\text{total}} = L_{MT} + \sum_{m=1}^M L_m \quad (7)$$

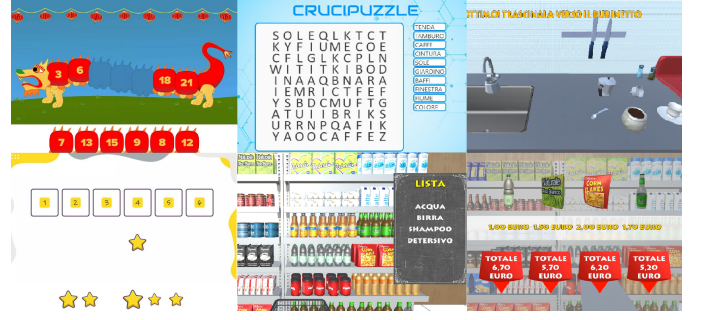


Fig. 2. Example screenshots from the cognitive games presented to the subjects. From the top left: *Complete the sequence*, *Word search*, *Coffee preparation*, *Order by size*, *At the grocery shop*, and *Bill payment*.

where L_{MT} is the categorical cross-entropy loss for the total MoCA score, and L_m is the loss for the m -th individual MoCA category. The cross-entropy loss for each L_m is defined as:

$$L_m = - \sum_{c=1}^C y_c^{(m)} \log(\hat{y}_c^{(m)}) \quad (8)$$

where C is the number of discrete classes for category m , $y_c^{(m)} \in \{0, 1\}$ is the one-hot encoded ground-truth label, and $\hat{y}_c^{(m)}$ is the predicted probability for class c . Regarding the computation of the MoCA categories loss, the ground-truth labels for each game are multiplied by a binary mask that enables the model to back-propagate error gradients only to the category classification heads that the game-related data is associated with.

This approach ensures that only meaningful visual information is processed, reducing noise and improving model interpretability. By leveraging the pretrained ViT and integrating the time-aware embeddings, the method effectively captures the spatial and temporal patterns of visual attention, making it a powerful tool for cognitive assessment. The resulting model (Fig. 1) offers an automated alternative to manual MoCA scoring, potentially enabling more efficient and objective cognitive screening in clinical and research settings.

IV. EVALUATION

A. Dataset acquisition

The study protocol involved the participation of 30 patients diagnosed with MCI, recruited from the Neuro-Geriatrics outpatient clinic at Cannizzaro Hospital in Catania. Each patient completed the MoCA questionnaire, with scores ranging from 14 to 29. The collected dataset includes a wide range of multi-modal data: eye-tracking data were recorded at a variable sample rate of 30-120 Hz to ensure temporal precision, capturing saccadic movements, fixations, and smooth tracking. These parameters are crucial for assessing patients' attention and visual behavior during cognitive exercises. Using the eye-tracking data, saliency maps were generated to highlight which regions of the visual interface in the cognitive games attracted the participants' attention most intensely. Screen recordings

TABLE I
GAME - MoCA ATTRIBUTE ASSOCIATIONS. VS: VISUO-SPATIAL, N:
NAMING, AT: ATTENTION, L: LANGUAGE, AB: ABSTRACTION,
DR: DELAYED RECALL, O: ORIENTATION

Game	VS	N	AT	L	AB	DR	O
Coffee preparation		✓				✓	✓
At the grocery shop		✓				✓	
Bill payment			✓				
Action and reaction		✓		✓		✓	
Word search			✓				
Order by size	✓		✓				
Complete the sequence	✓		✓		✓	✓	

documented patients’ interactions with the cognitive games, allowing an in-depth analysis of their choices and responses during cognitive tasks, including response time, answer accuracy, and action sequence. The cognitive training consisted of seven digital games, designed to assess and stimulate different cognitive and functional abilities. Each game was also mapped to specific cognitive functions assessed by the MoCA. The *Word search* and *Order by size* were associated with visuo-spatial and executive functions, while *Coffee preparation*, *At the grocery store*, and *Action and reaction* were linked to naming abilities. Attention was assessed through *Bill payment*, *Word search*, *Order by size*, and *Complete the sequence*, while *Action and reaction* also engaged language skills. Abstract reasoning was stimulated by *Complete the sequence*, and delayed recall was targeted in *Coffee preparation*, *At the grocery store*, *Action and reaction*, and *Complete the sequence*. Finally, *Coffee preparation* also contributed to orientation assessment. Table I reports the games and the associated MoCA attributes, while Fig. 2 shows example screenshots of the games.

Each patient participated in a single recording session lasting approximately one hour. Additionally, all participants signed an informed consent form, which included a detailed section on data collection and usage. Along with the multi-modal data, demographic information such as age, sex, and educational level was collected, along with clinical details including medical history and current medications. The study included 13 male and 17 female participants, with ages ranging from 59 to 87 years (mean age: 75.4). Educational levels varied among participants: 2 had no formal education, 5 completed elementary school, 6 attended lower secondary school, 10 completed upper secondary school, and 7 had a university degree. Cognitive and functional assessments, including previous evaluations and MoCA scores, were also documented to provide a comprehensive overview of each patient’s condition.

B. Experimental setting

For all experiments, the model was trained for 100 epochs using the AdamW optimizer [18] with a weight decay of 5×10^{-5} . As backbone, we used the ViT-B/16 Vision Transformer pretrained on ImageNet. The learning rate was set to $1e-4$ and the batch size was set to 16. The input images were resized from a 1920×1020 initial resolution to 224×224 , which is

TABLE II
PERFORMANCE COMPARISON FOR DIFFERENT MoCA-RELATED TASKS.

Task	Balanced accuracy(%)
Total MoCA Classification	83.52 ± 0.10
MoCA Category Classification	87.51 ± 0.69
Total MoCA + MoCA Category (Ours)	89.93 ± 0.12

TABLE III
PERFORMANCE COMPARISON WHEN DIFFERENT POSITIONAL EMBEDDING METHODS ARE USED.

Positional Embedding	Balanced accuracy(%)
FIXED	79.45 ± 0.36
LEARNABLE	85.72 ± 2.10
TAPE (Ours)	89.93 ± 0.12

the suitable input resolution for ViT patch embeddings. Due to the relatively small number of data samples ($N = 309$), 5-fold cross-validation ($N_{Fold} = 62$, except the last one that has 61 items) was used and the obtained performance is expressed in terms of average values and standard deviations across all folds.

C. Performance

We evaluated the performance of the proposed pipeline on the dataset presented in the previous subsection. Table II reports the model’s accuracy in predicting both the total MoCA score and the individual category scores. To highlight the importance of formulating the task as a classification problem at both levels, we report the results for the total and category-level predictions separately within the same table.

The results show that training the model with both the total MoCA score and individual category scores (89.93%) leads to better performance than training it solely on the total MoCA score (83.52%), with an improvement of nearly 6.5 percentage points. It also outperforms the model trained only on individual MoCA categories (87.51%) by about 1.5 percentage points. This suggests that while the primary contribution to classification performance comes from the individual MoCA categories, incorporating the total MoCA score improves performance.

In order to evaluate the impact of the time-aware positional embedding, we also conducted the same training experiments by replacing it with a learnable linear positional encoding [16] and a vanilla transformer (sine/cosine) positional embedding [15]. The results, shown in Table III, demonstrate that the model with the time-aware positional embedding outperforms, on average by several percentage points, the learnable and fixed positional embeddings.

Additionally, in Table IV, we report the results obtained by varying the granularity of the time unit length (t_w). The results indicate that the smaller the time unit (i.e., the shorter the temporal window of eye-gaze data it contains), the more accurate the model becomes. This is because, within a smaller time window, the sampled eye-gaze points are more temporally homogeneous. However, by reducing the time-unit length below 1 second does not provide a meaningful improvement

TABLE IV
PERFORMANCE COMPARISON USING DIFFERENT TIME UNIT DURATIONS.

t_w	Balanced accuracy(%)
0.5 sec	89.57 ± 0.17
1 sec (Ours)	89.93 ± 0.12
2 sec	85.72 ± 0.66
5 sec	84.18 ± 0.65

in accuracy while increasing the model’s complexity, leading to premature model overfitting.

V. CONCLUSIONS

This study presented an approach to automating MoCA score estimation using eye-gaze data and ViTs. By leveraging gaze-tracking technology and deep learning techniques, the proposed pipeline effectively captures spatial and temporal patterns of visual attention during cognitive tasks. The integration of a time-aware positional embedding mechanism further enhances the model’s ability to process inconsistent eye-gaze data, ensuring robust feature extraction and reliable predictions.

Experimental validation on a dataset collected from cognitive gaming sessions demonstrated the feasibility and effectiveness of this method. The results highlight its potential as a scalable, objective, and efficient alternative to traditional manual MoCA evaluations. This approach not only reduces the labor-intensive nature of manual scoring but also minimizes human error, paving the way for broader applications in clinical and remote cognitive screening.

Future work could explore expanding the dataset size, incorporating additional cognitive tasks, and refining the model architecture to further improve performance. Additionally, integrating multi-modal data such as speech or facial expressions could enhance the system’s predictive accuracy [19]. Overall, this research underscores the significant potential of combining eye-gaze tracking with advanced machine learning techniques for cognitive assessment, offering promising prospects for early detection and management of mild cognitive impairment and related conditions.

ACKNOWLEDGMENT

This work has been supported by Ministero dell’Università e della Ricerca (Italy), PON 2014-2020, project: “4Frailty”, n. ARS01_00345, CUP: E66C18000200005. Raffaele Mineo is a PhD student enrolled in the National PhD in Artificial Intelligence, XXXVII cycle, course on Health and life sciences, organized by Università Campus BioMedico di Roma.

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